# TEAM CONTRIBUTION & WORK DISTRIBUTION REPORT

**Course: DLE-AI-202 (Deep Learning), Cohort I 2026 | Track 1: Pure Research**

**Project:** *Do Register Tokens Regularize Vision Transformers Under Data Scarcity? A Controlled 12-Run Ablation*

**Repository:** github.com/shahin1717/vit-research | **Submission Date:** September 8, 2026

## 1. Executive Overview & Compliance Statement (§6)

Per Section 6 of the project specifications, this report documents the workload allocation, code ownership, and technical deliverables across our team. The project investigated the regularizing capacity of register tokens ($K \in \{0, 1, 4, 8\}$) under extreme data scarcity (~10k CIFAR-100 images). All five members contributed equally (**20.0% per member**) across theoretical formulation, PyTorch engineering, A100 GPU sweeps, statistical reduction, and manuscript drafting.

## 2. Responsibility Matrix & Effort Allocation

| Member Name | GitHub / Contact | Core Ownership Domain | Primary Assigned Codebase Modules | Effort |
|---|---|---|---|---|
| **Shahin Alakparov** | shahin1717 sahinalekperov5@gmail.com | Core Architecture & Training Lead | src/models/register_vit.py, scripts/train.py, scripts/eval.py | **20.0%** |
| **Gulnisa Abdurahmanli** | gulnisa gulnisa.abdurahmanli@gmail.com | Data Engineering & Multi-Budget Lead | src/data/cifar100_subset.py (100 & 300 imgs/class), src/data/__init__.py | **20.0%** |
| **Narmina Ibrahimova** | nnrmina narminaibrahimova2@gmail.com | Metrics & Interpretability Lead | src/models/attention_hook.py, src/metrics/entropy.py, outliers.py | **20.0%** |
| **Emil Ahmedli** | emilahmedli5 emilahmedli1905@gmail.com | Ablation Sweeps & Hardware Lead | scripts/run_sweep.sh, scripts/run_databudget_sweep.sh, configs/ | **20.0%** |
| **Rufet Dosteliyev** | rufetdosteliyev rufetdosteliyev@gmail.com | Ablation Analytics, Figures & Paper Lead | src/utils/aggregate_databudget.py, scripts/plot_*.py, paper/ | **20.0%** |
| **TOTALS** | **5 Team Members** | **Balanced End-to-End Pure Research** | **24/24 Ablation Runs Verified (100% Passing Tests)** | **100.0%** |

## 3. Detailed Technical Contributions per Member

- **Shahin Alakparov (20.0%):** Engineered RegisterVisionTransformer wrapping timm ViT-Tiny (d=192). Implemented prepend logic for learnable registers $R \in \Re^{K \times d}$ and output slicing. Developed the AMP training harness in scripts/train.py (AdamW, linear warmup + cosine annealing, gradient clipping, checkpoint serialization) and verified 59/59 tests.
- **Gulnisa Abdurahmanli (20.0%):** Built generalized StratifiedCIFAR100Subset in src/data/cifar100_subset.py, sampling both 100 and 300 samples/class (10k and 30k images) with deterministic generator seeds. Implemented balanced 9k/1k and 27k/3k train/val splits, Bicubic RandomResizedCrop(224), AutoAugment, and standard test set evaluation loader with pinned memory and multi-worker prefetching.
- **Narmina Ibrahimova (20.0%):** Authored ViTAttentionHookManager in src/models/attention_hook.py, non-invasively intercepting attention weights across all 12 MHSA layers with automatic CUDA memory clearing. Coded Shannon attention entropy $H(A^{(l)})$ in src/metrics/entropy.py, patch-norm $3\sigma$ outlier rate in src/metrics/outliers.py, and generalization gap metrics.
- **Emil Ahmedli (20.0%):** Led execution of all 24 ablation sweep runs across both data budgets (12 × 100pc and 12 × 300pc). Configured WireGuard VPN (team1.conf) and remote NVIDIA A100-SXM4 GPU cluster environment. Authored automated execution runners scripts/run_sweep.sh and scripts/run_databudget_sweep.sh with GPU cache clearing, supervising zero-OOM execution across 8+ hours.
- **Rufet Dosteliyev (20.0%):** Led multi-seed ablation analytics, statistical reduction, and scientific reporting across both 100pc and 300pc experimental suites. Built aggregation engines (src/utils/logger.py and src/utils/aggregate_databudget.py) producing sweep_summary.json and summary.json. Authored comparative visualizers (scripts/plot_metrics.py, scripts/plot_databudget_comparison.py), LaTeX tables, and manuscript Section 5.

## 4. Academic Integrity Attestation & Sign-Off

We certify that this report accurately reflects the collaborative division of labor for our final project. All members actively reviewed, validated, and approved the codebase, empirical findings, manuscript, and oral defense deck.

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| **Shahin Alakparov**<br>*Core Architecture Lead*<br>Status: **Verified & Signed** | **Gulnisa Abdurahmanli**<br>*Data Pipeline Lead*<br>Status: **Verified & Signed** | **Narmina Ibrahimova**<br>*Metrics & Hooks Lead*<br>Status: **Verified & Signed** |

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|---|---|
| **Emil Ahmedli**<br>*Ablation Sweeps & Hardware Lead*<br>Status: **Verified & Signed** | **Rufet Dosteliyev**<br>*Ablation Analytics, Figures & Paper Lead*<br>Status: **Verified & Signed** |